

Project Objective

The goal of this project is to transform raw transactional data into a professional Business Intelligence solution using **Excel, SQL, and Power BI**. The suite consists of 4 specialized dashboards designed to track sales performance, customer behavior, and regional growth.

Phase 1: Data Cleaning (Excel)

Before importing the data, extensive cleaning was performed in Excel to ensure data integrity:

- **Date Format Issue:** The raw dataset had inconsistent date formats (MM/DD/YYYY and DD/MM/YYYY). This was fixed using the 'Text to Columns' feature and the **DATEVALUE** function to standardize all entries into a unified Date format recognized by Power BI.
- **Data Validation:** Removed duplicate records and handled missing values (nulls) in the Profit and Sales columns to avoid calculation errors.
- **Standardization:** Cleaned category and city names to ensure there were no redundant entries due to spelling mistakes.

Phase 2: Backend Data Management (SQL)

SQL was used as the robust engine to structure and optimize the data:

- **Schema Design:** Created a database and imported cleaned Excel data with optimized data types (DECIMAL, DATE, VARCHAR).
- **Data Transformation:** Performed LEFT JOINS to consolidate multiple tables (Sales, Customers, Products).
- **Performance:** Created SQL Views and Indexes to provide Power BI with pre-processed, high-speed data.
- **Verification:** Used GROUP BY and SUM queries to validate KPI totals before visualization.

Phase 3: Detailed Dashboard Breakdown

Dashboard 1: Executive Sales Overview

- **Purpose:** Provides a high-level summary for stakeholders.
- **Key Features:** KPI cards for Total Sales (\$457K), Total Profit, and Quantity. It includes a monthly sales trend line chart to monitor growth.

Dashboard 2: Customer Analytics

- **Purpose:** Analyzes buyer behavior and segmentation.
- **Key Features:** Breakdown of sales by segment (Consumer, Corporate, Home Office) and a list of Top 10 Customers by revenue and profit.

Dashboard 3: Geographic & Regional Deep-Dive

- **Purpose:** Visualizes performance across different locations.
- **Key Features:** * **Filled Map:** A heatmap of the US states using a Teal-Orange gradient.

- **State-City Matrix:** A hierarchical table with drill-down capabilities.
- **Conditional Formatting:** Applied red backgrounds to negative profit values to quickly identify loss-making cities.

Dashboard 4: Product & Category Performance

- **Purpose:** In-depth analysis of the inventory and product lines.
- **Key Features:** Comparison between Furniture, Technology, and Office Supplies. It uses slicers for Region and Category to allow interactive filtering.

Technical Implementation

- **Data Modeling:** Established relationships between tables for seamless filtering.
- **DAX Measures:** Created custom measures for Total Sales, Total Profit, and Profit Margin.
- **Visual Aesthetics:** Used a consistent **Orange & Teal** color palette for a professional UI/UX.
- **Value Formatting:** Converted all large currency values into "**K**" (**Thousands**) for better readability.

Final Conclusion

- This project demonstrates a full data pipeline—from raw data cleaning in Excel to structured querying and advanced visualization in Power BI. It allows businesses to identify profitable regions and improve underperforming product categories efficiently.